IdiomX: A Bilingual Benchmark for Idiom Understanding, Generation, and Cross-Lingual Retrieval

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Abstract

Idiomatic expressions pose a significant challenge for natural language understanding due to their non-compositional semantics and context dependence. Existing datasets are limited in scale, diversity, or task coverage. To address this, we introduce IdiomX, a large-scale bilingual benchmark for idiom understanding built through a multi-stage pipeline combining lexical-resource collection, rule-based filtering, LLM-based enrichment, and validation. The resulting dataset contains 123,336 contextualized examples covering 14,986 English idioms. Each idiom is associated with multiple generated contexts, including both idiomatic and literal usages, enabling balanced evaluation of compositional and non-compositional language understanding.

IdiomX is constructed from publicly usable lexical resources, primarily WordNet and Kaikki.org, followed by normalization and idiom-focused filtering to retain globally meaningful expressions.

We then use OpenAI GPT-4.1-mini in a controlled structured-generation pipeline to enrich each idiom with contextual examples, canonical and surface-form information, English and Arabic meanings, translations, and additional semantic metadata. The enrichment stage is followed by schema validation, consistency checks, lexical alignment checks, anomaly filtering, and targeted manual inspection.

We benchmark IdiomX on three tasks: idiom detection, context-to-idiom prediction, and Arabic context-to-idiom retrieval. Experiments include classical baselines, transformer-based classifiers, dense retrieval models, hybrid generation–retrieval systems, and task-specific contrastive fine-tuning. Results show that transformer models substantially outperform lexical baselines in idiom detection, the proposed hybrid model achieves the strongest performance for context-to-idiom prediction, and fine-tuned multilingual E5 retrieval produces the best Task 3 results, reaching 64.86% Top-1 and 81.46% Top-5 accuracy in the upgraded cross-lingual retrieval setting.

These findings demonstrate that idiom understanding benefits from combining high-quality bilingual supervision with task-specific modeling. IdiomX provides a strong foundation for future work in figurative language understanding, multilingual semantic retrieval, and idiom-aware language modeling.

Keywords: idioms; figurative language; bilingual benchmark; cross-lingual retrieval; Arabic NLP; large language models

I. Introduction

Idiomatic expressions are a fundamental and challenging component of natural language [1], as their meanings are often non-compositional and cannot be directly inferred from the meanings of their individual words. This makes idioms particularly difficult for natural language processing systems across tasks such as semantic understanding, translation, retrieval, and generation.

For example, the phrase “kick the bucket” cannot be interpreted compositionally, while even more transparent idioms may still require strong contextual reasoning to distinguish idiomatic from literal usage [1], as shown in prior work on idiom detection and cross-lingual semantic modeling.

Despite their importance, existing idiom resources remain limited in one or more dimensions, including scale, contextual richness, annotation depth, multilingual coverage, and benchmarking support.

Many datasets contain only a small number of idiom types (e.g., verb–noun constructions, fixed expressions, or phrasal idioms), focus on a single language, or provide narrow annotations that are insufficient for modern deep learning and cross-lingual modeling.

To address these limitations, we present IdiomX, a large-scale bilingual benchmark for idiom understanding and modeling. IdiomX combines lexical-resource collection with controlled large language model (LLM) enrichment to produce contextualized idiom data with English–Arabic semantic alignment. The dataset is designed not only as a resource, but as a unified benchmark supporting multiple tasks that reflect different aspects of idiom understanding.

Each idiom in IdiomX is associated with multiple contextual examples covering both idiomatic and literal usages. This paired design enables precise evaluation of models' ability to distinguish figurative meaning from compositional interpretation, which remains a core challenge in figurative language processing.

A key motivation of this work is that idiom understanding should be studied across both monolingual and multilingual settings. In particular, mapping Arabic contextual descriptions to canonical English idioms defines a challenging retrieval task that emphasizes semantic alignment rather than surface similarity. This makes IdiomX suitable for evaluating both contextual language models and retrieval-based systems under realistic ambiguity.

Our contributions can be summarized as follows:

1. We introduce IdiomX, a large-scale dataset of idiomatic expressions with over 123K examples, enriched with contextual usage, bilingual meanings, and structured semantic annotations (balanced between idiomatic and literal usage) covering 14,986 idioms.
2. We propose a scalable data construction pipeline combining lexical-resource collection, LLM-based enrichment, structured validation, and targeted manual inspection.
3. We propose a unified benchmark covering multiple idioms understanding tasks, including idiom detection, contextual retrieval, cross-lingual mapping, normalization, and meaning-based retrieval.
4. We provide a comprehensive evaluation comparing classical machine learning models and transformer-based architectures for idiom detection.
5. We demonstrate that hybrid modeling improves context-to-idiom prediction, while task-specific contrastive fine-tuning significantly enhances cross-lingual retrieval performance, achieving up to 64.86% Top-1 and 81.46% Top-5 accuracy in Task 3.

Overall, IdiomX provides both a reusable dataset and a comprehensive benchmark for studying idiomatic language across modeling paradigms and languages.

II. Related Work

A. *Idiom Datasets*

Idiomatic expression understanding has been widely studied across several NLP tasks, including idiom detection, interpretation, and paraphrasing. Early work focused on lexical resources such as WordNet and Wiktionary, which provide structured definitions and semantic relations but lack contextual diversity and large-scale annotations.

For example, in the VNC-Tokens dataset, the phrase "take place" may be labeled as idiomatic when meaning "to occur", but literal in contexts describing physical movement. Similarly, SemEval-2013 Task 5b [10] includes expressions such as "make up", which can denote either "to invent" or "to compose", depending on context.

Unlike MAGPIE [8], which focuses on potentially idiomatic expressions without controlled contextual generation, IdiomX provides balanced idiomatic and literal examples with explicit supervision. Similarly, compared to SemEval-2013 Task 5b [10], which is limited in size, IdiomX offers significantly larger scale and richer annotations.

B. *Neural Models for Idiom Understanding*

Recent work has explored neural approaches to idiom understanding using contextual language models and sequence-to-sequence architectures. Transformer-based models such as BERT [2] have been widely used for idiom detection and classification, where the objective is to distinguish figurative from literal usage in context. Sequence-to-sequence models such as T5 [7] have also been applied to idiom interpretation, paraphrasing, and generation tasks, benefiting from their ability to model semantic reformulation.

For instance, transformer-based models such as BERT [2] classify sentences like "He kicked the bucket" as idiomatic, while sequence-to-sequence models such as T5 [7] can generate paraphrases such as "He passed away". These examples highlight the importance of contextual modeling in idiom understanding.

These approaches have achieved strong results when sufficient contextual supervision is available. However, their effectiveness remains closely tied to dataset quality, scale, and diversity. Cross-lingual idiom understanding remains much less studied than monolingual detection or interpretation, and retrieval-oriented settings are still underrepresented.

This motivates the need for a benchmark that supports not only idiom classification, but also generation and cross-lingual retrieval under realistic contextual ambiguity.

To better position IdiomX within this landscape, Table 1 provides a quantitative comparison with widely used idiom datasets.

Dataset	Lang	Idioms	Inst.	Bi	Ref
VNC-Tokens	EN	53	2,984	No	ACL Anthology
SemEval-2013 Task 5b [10]	EN	85	~4,350	No	ACL Anthology
IDIX [9]	EN	52	4,022	No	ACL Anthology
MAGPIE [8]	EN	~1,700	56,000	No	ACL Anthology
PIE-English [11]	EN	~1,200	~20,100	No	arXiv
IdiomX (Ours)	EN+AR	14,986	123,336	Yes	Public Repository [1]

Table 1: Comparison of IdiomX with existing idiom datasets

This comparison highlights the gap between existing monolingual datasets and the need for large-scale bilingual resources with structured semantic annotations.

Beyond scale, IdiomX distinguishes itself through its balanced construction of idiomatic and literal examples, as well as its rich linguistic annotations, enabling deeper analysis of

[1] Public resources are available in the dataset and project repositories

ambiguity, compositionality, and semantic consistency. In addition, IdiomX integrates controlled LLM-based enrichment to generate diverse contextual examples and bilingual semantic representations, supporting a broader range of tasks including detection, generation, and cross-lingual retrieval.

III. DATASET CONSTRUCTION

IdiomX is a large-scale dataset of English idiomatic expressions enriched with structured semantic annotations, bilingual meanings, and contextual usage examples. The dataset is constructed from multiple lexical resources, primarily Kaikki.org and WordNet, followed by filtering, enrichment, and validation stages designed to maximize precision and support downstream modeling.

A. Data Collection and Filtering

The IdiomX dataset is constructed from multiple lexical resources, including Kaikki.org (a structured Wiktionary-derived dataset) and WordNet, selected for their coverage and open licensing.

We first extract candidate multi-word expressions from the Kaikki dataset, resulting in over 229,000 entries. To ensure high precision, a strict filtering stage is applied to remove compositional phrases, incomplete entries, and non-idiomatic expressions. This process reduces the dataset to 30,749 candidate idioms.

A subsequent cleaning and normalization stage standardizes text formatting, removes duplicates, and handles missing values, yielding 21,816 entries. To further improve quality, we construct a high-precision subset using linguistic heuristics such as idiomatic tags, figurative meaning indicators, and known idiomatic patterns, resulting in 13,202 high-confidence idioms.

In parallel, multi-word expressions are extracted from WordNet. While WordNet provides broader lexical coverage, many extracted expressions are compositional rather than idiomatic. These entries are therefore treated as lower-confidence candidates and filtered more aggressively in later stages.

B. Dataset Integration and Statistics

The high-precision Kaikki dataset is merged with the WordNet-derived expressions to increase coverage. A global filtering stage is then applied to remove weak or compositional expressions, particularly from WordNet.

The final dataset contains 12,853 high-quality idioms, with 12,847 unique expressions. Among these, 8,891 entries include example sentences, while 3,962 lack contextual usage.

To ensure consistency across sources, all entries are normalized into a unified schema containing fields such as idiom text, English meaning, example sentence, part-of-speech, source, and confidence level.

This curated dataset forms the foundation for subsequent enrichment and modeling tasks.

C. LLM-Based Enrichment

To enrich the dataset with contextual usage and semantic annotations, we employ a large language model (LLM)-based generation pipeline.

We employ GPT-4.1-mini due to its favorable trade-off between generation quality and computational scalability, enabling large-scale controlled generation while maintaining semantic consistency.

Each idiom is processed using a structured prompting strategy that enforces a fixed JSON output schema.

For each idiom, the model generates:

- multiple contextual examples covering both idiomatic and literal usage,
- canonical and surface forms,
- English and Arabic meanings,
- and linguistic attributes such as ambiguity, compositionality, domain, and learner difficulty.

A key design choice is the generation of paired idiomatic and literal examples for the same expression. This allows the same surface form to appear in both figurative and literal contexts, enabling controlled evaluation of semantic understanding.

Although part of the dataset is synthetically generated, multiple validation stages ensure linguistic plausibility and semantic correctness, mitigating common issues associated with synthetic data generation.

For example, for the idiom "kick the bucket", the model generates:
- **Idiomatic**: "*He finally kicked the bucket after a long illness.*"
- **Literal**: "*He kicked the bucket across the yard.*"

The use of structured JSON outputs ensures reliable parsing, consistency, and reproducibility across large-scale generation.

One limitation of this approach is the reliance on LLM-generated examples, which may introduce subtle biases or unnatural constructions despite validation.

D. **Validation and Quality Control**

To ensure data quality, we apply a multi-stage validation pipeline combining rule-based checks and LLM-assisted verification.

Rule-based validation enforces structural and semantic consistency through checks on:

- idiom presence in the generated sentence,
- validity of usage labels (idiomatic vs. literal),
- missing or malformed fields,
- and minimum length and quality constraints.

Records that fail validation are flagged for further inspection. Instead of discarding such entries, we apply a secondary LLM-based verification step to correct errors and improve consistency.

Each record is assigned a validation status (valid, corrected, or needs review), resulting in:

- 121,079 valid entries,

- 1,531 verified entries,
- 726 corrected entries.

The final dataset consists of over 123,000 examples with rich annotations and bilingual coverage. Figure 1 summarizes the overall construction pipeline, after which we analyze the resulting dataset properties in greater detail.

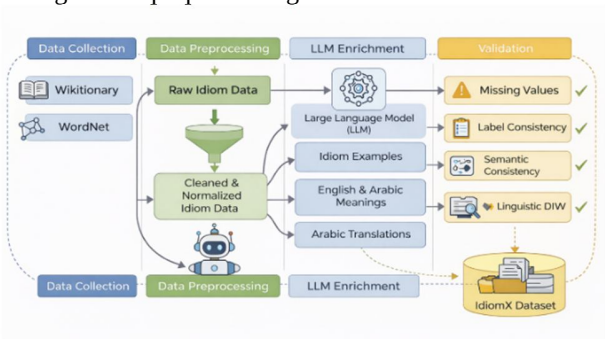

Figure 1: Overview of the IdiomX dataset construction pipeline

Additional semantic metadata is analyzed to characterize dataset properties and ensure diversity across linguistic dimensions.

In terms of compositionality, idioms are distributed across semi-opaque (71,296), opaque (29,472), and transparent (22,568) categories, reflecting varying degrees of semantic non-compositionality. Ambiguity analysis shows that 80,904 examples allow partial literal interpretation, while 30,720 are strongly idiomatic.

Learner difficulty annotations further indicate that most idioms fall within the medium difficulty range (89,528), with smaller proportions classified as easy (17,552) and hard (16,256).

These distributions demonstrate that the dataset captures a wide spectrum of idiomatic behavior, ambiguity, and semantic complexity, supporting robust evaluation across multiple modeling tasks.

E. Dataset Analysis and Distribution

To better understand the characteristics of the IdiomX dataset, we analyze the distribution of key linguistic and structural features, focusing on label balance, compositionality, ambiguity, and semantic quality.

Figure 2 illustrates the distribution of idiomatic versus literal examples, confirming the effectiveness of the controlled generation strategy and the near-perfect class balance of the dataset. Figures 3 and 4 further characterize ambiguity and compositionality. Together, these distributions show that the dataset includes a wide range of semi-literal, opaque, and transparent cases, reflecting realistic scenarios in which idiomatic and compositional interpretations may overlap. For example, partially compositional idioms such as “spill the beans” retain some connection to their literal meaning while still requiring contextual interpretation.

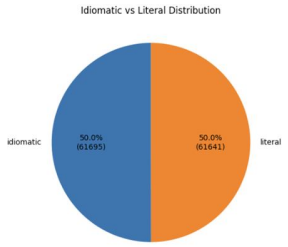

Figure 2: Idiomatic vs Literal Distribution

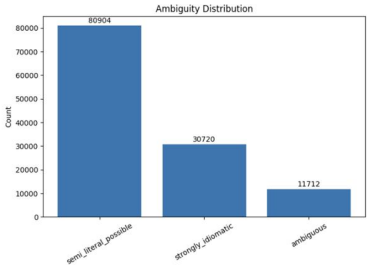

Figure 3: Ambiguity Distribution

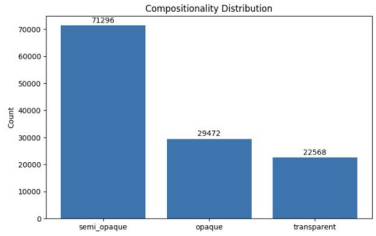

Figure 4: Compositionality Distribution

As shown, the dataset exhibits a wide range of ambiguity levels, with a large proportion of semi-literal cases, reflecting realistic linguistic scenarios where idiomatic meaning may overlap with compositional interpretation. Similarly, the compositionality distribution spans semi-opaque, opaque, and transparent idioms, indicating that the dataset captures varying degrees of semantic complexity.

For example, partially compositional idioms such as "spill the beans" retain some connection to their literal meaning while still requiring contextual interpretation.

Figures 5 and 6 present the distributions of example length, semantic consistency, and canonical semantic scores. These distributions appear smooth and well-structured, suggesting stable variation across examples and consistent annotation quality.

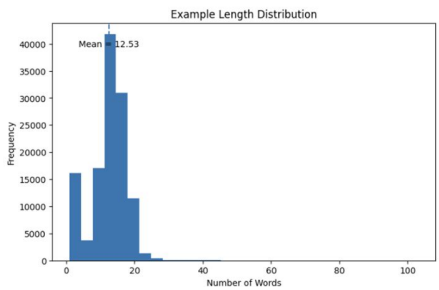

Figure 5: Example Length Distribution

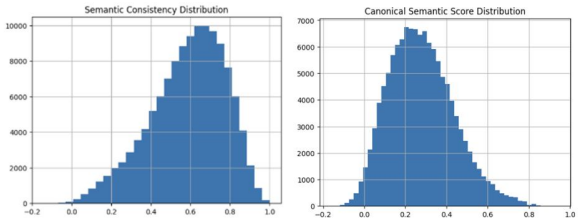

Figure 6: Semantic Consistency and Canonical Semantic Distribution

Finally, Figures 7, 8, and 9 provide a deeper view of semantic quality across labels, learner difficulty levels, and domains.

Figure 7 shows that idiomatic examples exhibit lower semantic similarity to their literal forms but maintain strong internal consistency, confirming that the dataset effectively distinguishes between figurative and compositional usage.

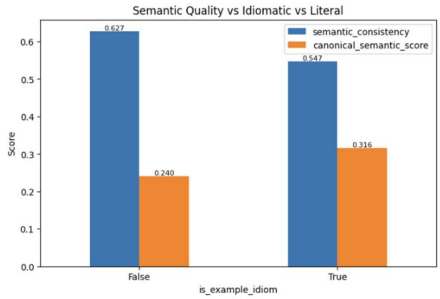

Figure 7: Semantic Quality vs Label

Figure 8 highlights variation across learner difficulty levels, where more difficult idioms tend to exhibit lower semantic similarity, reflecting increased abstraction and reduced compositional transparency.

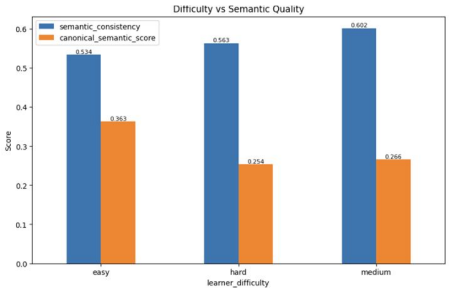

Figure 8: Difficulty vs Semantic Quality

Figure 9 illustrates the domain distribution, showing that while the dataset is dominated by general-domain idioms, it also includes a diverse range of specialized domains, supporting broader applicability across NLP tasks.

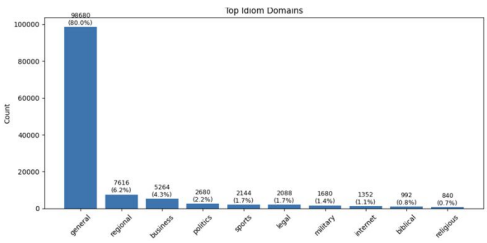

Figure 9: Domain Distribution

IV. Benchmark Tasks

IdiomX supports multiple downstream NLP tasks involving idiomatic language.

A. *Task 1: Idiom Detection.*

B. *Task 2: Context-to-Idiom Prediction.*

C. *Task 3: Arabic Context-to-Idiom Retrieval.*

Similar tasks have been explored in prior work on idiom understanding and figurative language processing.

Idiom detection has been studied as a classification problem (e.g., SemEval-2013 Task 5b [10]), while context-to-idiom prediction relates to lexical substitution and paraphrase generation tasks.

Cross-lingual idiom retrieval remains less explored but is related to multilingual semantic retrieval and cross-lingual sentence matching using transformer-based models.

These benchmark tasks are made possible by the structured design of IdiomX. Specifically, the dataset includes fields such as:

- The input sentence (idiom_in_example),
- binary usage labels (is_example_idiom),

And aligned semantic representations in both English and Arabic:

  - idiom_in_example_meaning_en,
  - idiom_in_example_meaning_arabic.

These fields were explicitly introduced during dataset construction to enable supervised learning, evaluation, and cross-lingual benchmarking across multiple idiom understanding tasks.

V. Experiments

To evaluate the effectiveness of IdiomX, we conduct baseline experiments across multiple tasks supported by the dataset.

A. *Task 1: Idiom Detection*

Idiom detection is formulated as a binary classification task [1], where the objective is to determine whether a given sentence expresses an idiomatic meaning or a literal interpretation.

This task plays a critical role in natural language understanding, as idiomatic expressions often convey meanings that cannot be inferred directly from individual words [1]. Accurate detection is essential for applications such as:

- Machine translation,
- Search systems,
- Conversational agents,
- Large language model (LLM) evaluation, and
- Language Learning Tools.

Unlike standard text classification, idiom detection requires distinguishing between surface-level lexical patterns and deeper contextual meaning, making it a challenging semantic task [1].

1. *Pipeline Overview*

The idiom detection pipeline consists of three main stages:

(1) Data Preparation, Preprocessing and Split Construction

The benchmark is constructed from the IdiomX dataset, where each instance includes an input sentence and a binary label indicating literal or idiomatic usage. The data is cleaned by removing invalid entries, filtering labels, and eliminating duplicates. A balanced splitting strategy is applied to ensure fair evaluation across training, validation, and test sets.

(2) Feature Representation

Different feature representations are used depending on the model type. For classical models, TF–IDF features with uni-grams and bi-grams are extracted to capture surface-level lexical patterns. For transformer-based models, tokenized sentences are processed using pretrained language models [2][3] to obtain contextual embeddings.

(3) Model Training and Evaluation

Models are trained using supervised learning. Logistic Regression is trained on TF–IDF features, while RoBERTa [3] is fine-tuned end-to-end using cross-entropy loss. All experiments are conducted under consistent preprocessing and evaluation settings to ensure reproducibility.

2. *Models*

We formulate idiom detection as a binary classification task. Given an input sentence $x = (w_1, w_2, \ldots, w_n)$, where each w_i represents a word token, the model predicts a label $y \in \{0,1\}$, where $y = 1$ indicates idiomatic usage and $y = 0$ indicates literal usage.

We evaluate both classical machine learning models and transformer-based architectures.

For classical models, we use Logistic Regression with TF–IDF features (uni-grams and bi-grams). The model estimates the probability of idiomatic usage as:

$$P(y = 1 \mid x) = \sigma(w^T x + b)$$

where σ is the sigmoid function. The model is trained using binary cross-entropy loss and primarily captures surface-level lexical patterns.

For transformer-based models, we use RoBERTa [3], which computes contextual representations using self-attention mechanisms [2]. The final prediction is obtained from the [CLS] token representation:

$$\hat{y} = \mathrm{softmax}(W\, h_cls + b)$$

This allows the model to capture contextual meaning beyond individual word occurrences.

We evaluate the following models:
- Logistic Regression (baseline)
- RoBERTa (base)

3. *Results*

We compare the performance of surface-level lexical models and deep contextual representations.

This task is aligned with prior contextual idiom classification work [12].

As shown in Table 2, RoBERTa consistently outperforms Logistic Regression across all evaluation metrics.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.8536	0.8476	0.862	0.855
RoBERTa [3]	0.9137	0.9056	0.924	0.914

Table 2: Performance comparison on the idiom detection task

Accuracy = (TP + TN) / (TP + TN + FP + FN)
Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1 Score = 2 * (Precision * Recall) / (Precision + Recall)

The improvement is particularly noticeable in recall, indicating that RoBERTa [3] is more effective at identifying idiomatic expressions.

Performance differences between models are consistent across validation and test splits.

As illustrated in Figure 10, transformer-based models achieve higher performance across all metrics [2][3], with the largest improvement observed in recall.

This suggests that contextual models are better suited to handling ambiguous expressions [1][2], where meaning cannot be inferred from individual words alone.

For example, the sentence “spill the beans about the plan” is correctly classified as idiomatic by RoBERTa [3], while Logistic Regression may misclassify it due to reliance on surface-level lexical features.

Overall, the visual comparison (Figure 10) confirms that contextual representations improve model performance, particularly in challenging cases involving semantic ambiguity.

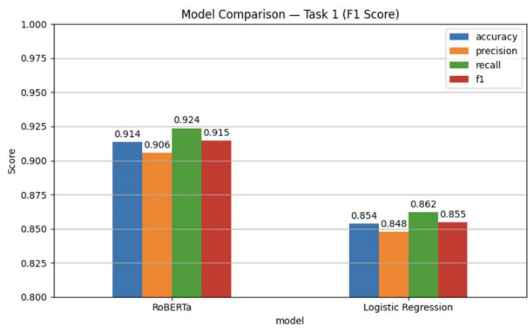

Figure 10: Model comparison on the idiom detection task (F1 score)

4. *Calibration and Confidence*

To further analyze model behavior, we examine confusion matrices, ROC curves, precision–recall curves, confidence distributions, and probability calibration [4].

The confusion matrices (Figure 11) show that RoBERTa [3] reduces both false positives and false negatives compared to Logistic Regression, indicating more balanced performance

across classes.

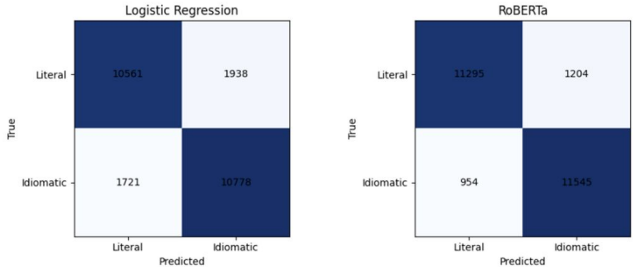

Figure 11: Confusion matrices comparison

The ROC curve (Figure 12) shows that RoBERTa [3] achieves a higher AUC (0.973) compared to Logistic Regression (0.926), which serves as our baseline model, demonstrating stronger discriminative ability across decision thresholds.

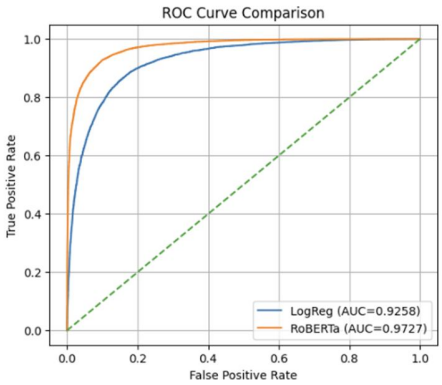

Figure 12: Roc Curve Comparison

Similarly, the precision–recall curve (Figure 13) indicates that RoBERTa [3] maintains higher precision across recall levels, confirming its robustness in identifying idiomatic expressions.

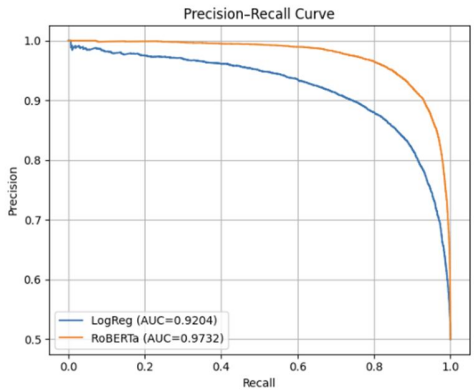

Figure 13: Precision-Recall Curve Comparison

The confidence distribution (Figure 14) shows that RoBERTa [3] produces highly confident predictions for correct classifications, while incorrect predictions are generally associated with lower confidence.

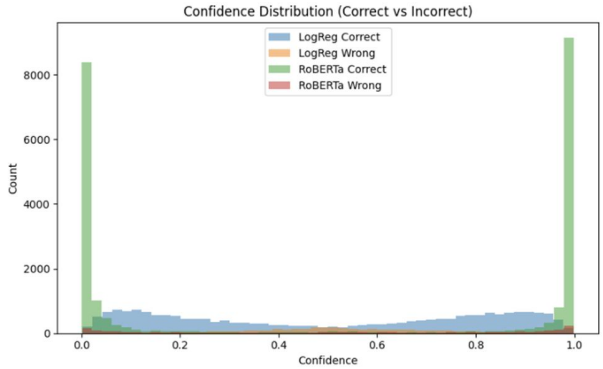

Figure 14: Confidence Distribution Comparison (Correct vs Incorrect)

For example:
- Correct prediction (high confidence):
"He finally spilled the beans about the plan" → idiomatic (confidence ≈ 0.99)
- Incorrect prediction (lower confidence):
Short or ambiguous phrases such as "baby sister" may produce lower confidence and occasional misclassification.

In contrast, Logistic Regression exhibits greater overlap between correct and incorrect confidence values, indicating less reliable probability estimates.

Overall, RoBERTa [3] provides more meaningful and better-calibrated confidence scores.

The calibration [4] curve (Figure 15) shows how well the predicted probabilities match the true outcomes. The dashed diagonal line represents perfect calibration [4].

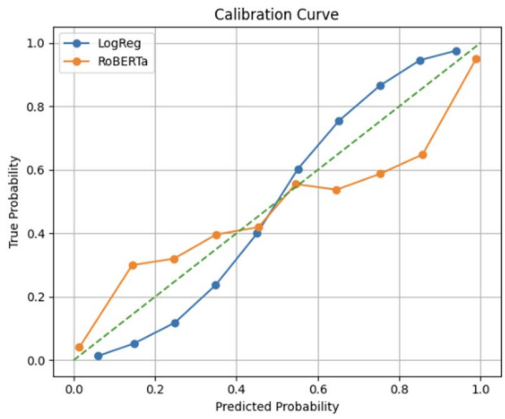

Figure 15: Calibration Curve Comparison

RoBERTa [3] stays closer to this ideal line across most probability ranges, meaning its predicted confidence aligns better with actual correctness.

Logistic Regression deviates more from the diagonal, especially at higher confidence levels, indicating that it can be overconfident or less accurate in its probability estimates.

This result demonstrates that transformer models not only improve classification performance but also provide more trustworthy probability estimates [4], which is important for real-world decision-making systems.

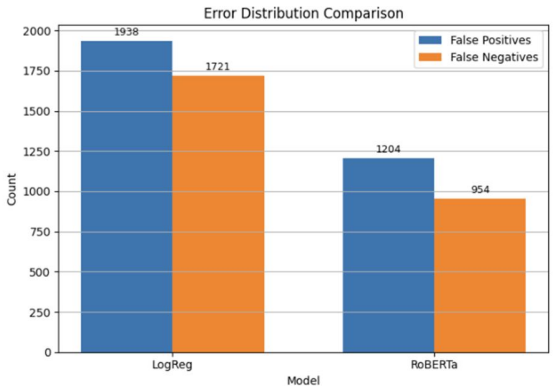

Figure 16: Error count comparison

Qualitative Example of Contextual Disambiguation

To further illustrate model behavior, we present a representative example demonstrating RoBERTa's [3] ability to distinguish between idiomatic and literal usage based on context.

Sentence 1:
"*He finally kicked the bucket after a long illness.*"
Prediction: **Idiomatic**.
Probability (idiomatic): 0.9973.

Sentence 2:
"*She kicked the bucket across the yard.*"
Prediction: **Literal**.
Probability (idiomatic): 0.0066.

This example highlights that the model correctly identifies the figurative meaning in the first sentence while recognizing the literal action in the second. The high confidence in both predictions indicates strong contextual understanding and reliable decision-making.

This demonstrates that the model is able to rely on contextual cues rather than surface-level word matching, correctly distinguishing between identical phrases used in different meanings.

5. *Error Analysis and Key Failure Modes*

(1) Short or context-limited inputs
Short phrases such as "baby sister" or "Cuban bread" lack sufficient context, making it difficult to determine whether the usage is literal or idiomatic.

(2) Named entities and domain-specific expressions
Phrases such as "Benjamin Button" or "Bravo Zulu" may be misclassified due to ambiguity between literal and idiomatic interpretation.

(3) Context-dependent idioms
Expressions like "one for the road" require broader discourse context, which may not be fully captured in single-sentence inputs.

(4) Ambiguous phrases
Certain expressions can be interpreted both literally and idiomatically depending on context, representing inherent challenges in the task.

Tables 3 and 4 present representative examples of false negatives (missed idioms) and false positives (literal sentences misclassified as idiomatic), respectively.

Sentence	True Label	Idiomatic Probability
The hikers waited out the night in the tent during the heavy rainfall.	Idiomatic	0.017
After fixing the machine quickly, he could have a good time of it for the rest of the day.	Idiomatic	0.056
The soldiers were taught to duck down quickly to evade enemy fire.	Idiomatic	0.08
I won't wear new shoes tomorrow because my dogs are barking today.	Idiomatic	0.016
He said he would help, but only when two Sundays meet.	Idiomatic	0.476

Table 3: Examples of false negatives.

Table 4 examples of false positives (literal sentences misclassified as idiomatic).

Sentence	True Label	Idiomatic Probability
Close their eyes and think of England.	Literal	0.527
The pairing of the spicy chili and the salty cheese was a match made in hell for my sensitive stomach.	Literal	0.692
She chose to take his words at face value, believing every promise without looking for hidden meanings.	Literal	0.998
A true friend of the environment would take steps to reduce waste and pollution.	Literal	0.928
He needed some solitude to rest his soul after hearing the sad news.	Literal	0.766

Table 4: Examples of false positives.

These examples show that most errors occur in sentences where literal and idiomatic meanings overlap or where contextual cues are limited.

False negatives typically involve idioms expressed in subtle or less explicit forms, while false positives often arise when literal sentences contain words commonly associated with idiomatic expressions.

This highlights that remaining errors are primarily due to semantic ambiguity [1] rather than simple lexical confusion.

6. *Summary*

Overall, the results suggest that idiom detection benefits from contextual understanding.

While Logistic Regression provides a strong baseline, its reliance on surface lexical features limits its ability to handle ambiguous expressions. In contrast, RoBERTa [3] achieves higher accuracy and more balanced error distribution by leveraging contextual representations.

However, remaining errors highlight the inherent ambiguity of idiomatic language and the importance of richer contextual and semantic understanding in future work.

B. *Task 2: Context-to-Idiom Generation*

In this task, the model is given a contextual sentence or semantic description and is required to predict the most appropriate idiomatic expression. Unlike idiom detection, this task requires both semantic understanding and generation capability, making it more challenging and closer to real-world language use.

Formally: $y^* = \text{argmax}_y P(y \mid x)$
where y is selected from a predefined idiom inventory.

A. ***SBERT Retrieval Model***

Each context x and candidate idiom i are encoded into vector representations:

$h_x = \text{Encoder}(x)$, $h_i = \text{Encoder}(i)$
Similarity is computed using cosine similarity [5]:
$\text{sim}(x, i) = (h_x \cdot h_i) / (\|h_x\| \, \|h_i\|)$
Prediction: $\hat{y} = \text{argmax}_i \, \text{sim}(x, i)$

B. ***FLAN-T5 Generative Model***

The model generates an idiom sequence y autoregressively [6]:

$P(y \mid x) = \prod P(y_t \mid y_{<t}, x)$
Prediction: $\hat{y} = \text{argmax}_y \, P(y \mid x)$

C. ***Proposed FLAN-First Hybrid Model***

"$\hat{y}$ equals y_gen if Match $\geq \tau$, otherwise the best retrieval candidate."

$\text{Match}(y_gen, I) = \max_i \, \text{similarity}(y_gen, i)$

1. *Experimental setup*

$$\text{Top-K Accuracy} = (1/N) \sum \mathbb{1}\{y \in \text{TopK}(x)\}$$
$$\text{MRR} = (1/N) \sum (1 / \text{rank}_i)$$

2. *Quantitative Results:*

Model	Exact Match (Top-1)	Top-3	Top-5	MRR@100
SBERT Retrieval	0.461	0.599	0.65	0.549
FLAN-T5 [6]	0.678	-	-	-
Hybrid FLAN-first	0.818	-	-	-

Table 5: Models exact Match - Top-1, Top-2, Top-3.

Performance differences between models are consistent across validation splits.

The results show that:

SBERT [5] provides strong semantic retrieval but suffers from ranking limitations.

FLAN-T5 [6] improves contextual reasoning but may produce invalid or noisy outputs.

The hybrid model combines generative reasoning with retrieval grounding, improving both accuracy and robustness.

This confirms that idiom prediction benefits from a two-stage reasoning process, where generation captures semantic intent and retrieval enforces discrete correctness.

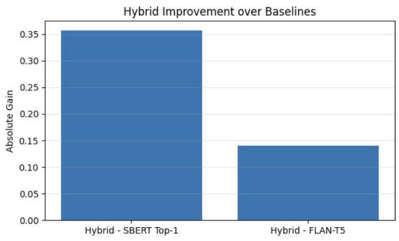

Figure 9: Hybrid Improvement over Baselines

Comparison	Gain
Hybrid vs SBERT [5]	0.356
Hybrid vs FLAN [6]	0.14

Table 6: comparison table gain from hybrid to SBERT and FLAN.

The hybrid model improves exact-match accuracy by 35.7 percentage points over SBERT [5] Top-1 and by 14.0 points over FLAN-T5 [6].

Decision	Count	Percent
FLAN exact bank	7885	0.724458
FLAN fuzzy correction	608	0.055862
SBERT [5] fallback	2391	0.21968

Table 7: comparison table hybrid Decision Distribution.

3. *Hybrid Decision Analysis*

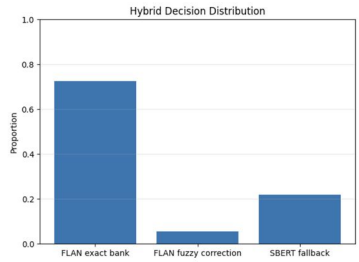

Figure 10: Hybrid Decision Distribution

Most predictions are resolved by the generative model, while retrieval provides corrective support.

4. *Error Analysis*

Errors are categorized into:
1. Retrieval Error: correct idiom not retrieved
2. Ranking Error: correct idiom not ranked first
3. Generation Error: incorrect or noisy output

The hybrid model reduces both ranking and generation errors

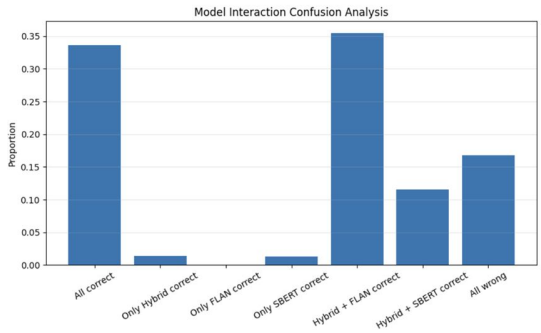

Figure 11: Model Interaction Confusion chart

These results indicate that the hybrid model is not merely averaging predictions but recovering additional correct cases through complementary decision behavior.

5. *Qualitative Analysis*

Case Study 1: Hybrid corrects both models

- SBERT [5] retrieval fails to select the correct idiom
- FLAN [6] generates an incorrect or approximate phrase
- The hybrid model successfully corrects the output

Case Study 2: FLAN succeeds alone

- FLAN [6] correctly generates the idiom directly
- SBERT [5] retrieval may rank incorrect candidates

- The hybrid model accepts the FLAN output without modification

Case Study 3: SBERT [5] fallback saves FLAN [6]

- FLAN [6] produces incorrect or incomplete outputs
- SBERT [5] retrieval successfully identifies the correct idiom
- The hybrid model switches to retrieval-based prediction

These examples demonstrate complementary strengths between generation and retrieval.

6. Discussion

The results indicate that idiom prediction is not purely a retrieval task, but benefits from generative reasoning.

The hybrid model effectively integrates both paradigms, improving robustness and accuracy.

The hybrid approach introduces additional computational cost due to dual processing, but this is justified by the performance gain.

This improvement arises because generative models capture semantic flexibility, while retrieval models enforce discrete correctness within a constrained idiom inventory

7. Task Summary

The proposed hybrid model achieves the best performance on the context-to-idiom task.

By combining generative reasoning with retrieval-based grounding, it overcomes the limitations of both approaches and provides a robust solution for idiomatic understanding.

C. Task 3: Arabic Context-to-Idiom (Retrieval Benchmark)

In Task 3, the objective is to retrieve the canonical English idiom corresponding to an Arabic contextual sentence. Unlike Task 2, which is framed as a generation task, Task 3 is formulated as a cross-lingual semantic retrieval problem. The input is the Arabic idiomatic context, and the target is the associated canonical English idiom from a predefined inventory.

The task is evaluated as a ranking problem, where the correct idiom must be retrieved from a predefined candidate set.

We formulate the Arabic context-to-idiom task as a semantic retrieval problem rather than a generative task. This design choice is motivated by the fixed and canonical nature of idioms, where exact canonical matching is required at evaluation time. Retrieval-based approaches allow mapping Arabic contextual meaning into a structured embedding space and retrieving the most semantically similar idiom from a fixed candidate set.

1. Formulation

Each Arabic query x and candidate idiom i are mapped into vector representations:

$h_x = \text{Encoder}(x)$, $h_i = \text{Encoder}(i)$
Similarity is computed using cosine similarity [5]:
$\text{sim}(x, i) = (h_x \cdot h_i) / (\|h_x\| \|h_i\|)$

Prediction is defined as:
$\hat{y} = \text{argmax}_i \text{sim}(x, i)$

where sim(·) denotes cosine similarity [5] in a shared embedding space.

2. Experimental Setup

To construct the Task 3 benchmark, Task 3 was initially explored through several generation-based, hybrid, and retrieval-based scripts, including mT5 baselines, hybrid mT5–retrieval variants, and Arabic-bank retrieval models. Early experiments used a broad Arabic input formulation that combined multiple Arabic fields and adopted a strict split strategy designed to avoid overlapping at the idiom or canonical target level. While this setup was methodologically strict, it produced **near-zero retrieval accuracy**, indicating that the benchmark was too difficult in its initial form and poorly matched to the intended retrieval setting.

To address this, the task was reformulated as an example-level cross-lingual retrieval problem.

The final benchmark retained only idiomatic examples, used `idiom_in_example_arabic` as the Arabic query, and used `idiom_canonical` as the English retrieval target.

Instead of splitting by idiom identity, the data was split by example within each idiom, allowing the same idiom to appear in both train and test through different contextual realizations.

This setup evaluates whether the model can retrieve the correct idiom for a novel Arabic context while preserving a realistic candidate inventory.

Under this revised formulation, the final retrieval pipeline showed a clear progression with fine-tuned E5 (v2) where Top-1 = 64.86%, Top-5 = 81.46%

These results show that stronger pretrained multilingual embeddings provide only modest gains by themselves, whereas task-specific contrastive fine-tuning is the dominant factor behind performance improvement. Additional gains are obtained through richer supervision, such as using multiple positives per query and longer training. The final best-performing Task 3 system is therefore the upgraded fine-tuned multilingual E5 retrieval model.

3. Arabic Semantic Bank Construction

A semantic bank is constructed from the training split, where each idiom is represented through multiple Arabic textual views:

- Example sentences (idiom usage)
- Example meanings
- Canonical meanings

Each entry is encoded into a multilingual embedding space, resulting in a dense representation matrix. This multi-view representation improves semantic coverage and allows the model to match queries against diverse contextual signals.

4. Models Evaluated

We evaluate a range of multilingual retrieval models, including both baseline and fine-tuned configurations:

- Multilingual MiniLM (baseline)
- Multilingual E5 (base, zero-shot)
- Multilingual E5 (fine-tuned v1)
- Multilingual E5 (fine-tuned v2 with enhanced positives)

While all models are evaluated, detailed analysis focuses on the strongest-performing configurations.

5. *Quantitative Results*

Evaluation is conducted using Top-K accuracy metrics:
Top-K Accuracy = $(1/N) \sum \mathbb{1}\{y \in \text{TopK}(x)\}$

Model	Top-1	Top-5
Zero-shot MiniLM baseline	23.27%	31.79%
Zero-shot multilingual E5	28.16%	39.08%
Fine-tuned E5 v1	60.37%	76.07%
Fine-tuned E5 v2	64.86%	81.46%

Table 8: Models Accuracy comparison table.

The multilingual E5 model consistently outperformed the MiniLM baseline. Top-1 accuracy improved from 23.27% to 27.46%, while Top-5 accuracy increased from 31.79% to 38.34%. These gains indicate that instruction-aware multilingual retrieval embeddings provide stronger semantic alignment between Arabic contextual inputs and canonical English idiom candidates.

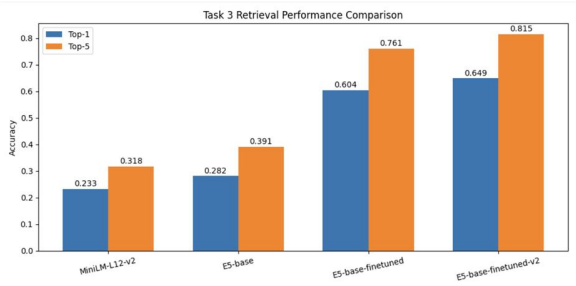

Figure 12: Model Retrieval Performance Comparison.

The results show a substantial improvement from fine-tuning, with E5-base-finetuned v2 outperforming all other models.

6. *Recall Analysis*

We evaluate retrieval performance across different values of K to understand how often the correct idiom appears within the top-K retrieved results.

The recall curve shows that:

- Recall increases rapidly for small values of K.
- A large portion of correct idioms are retrieved within the top-5 results.
- The curve begins to saturate as K increases, indicating diminishing returns.

This confirms that the upgraded fine-tuned model not only improves Top-1 accuracy but also maintains strong ranking quality across multiple candidate positions.

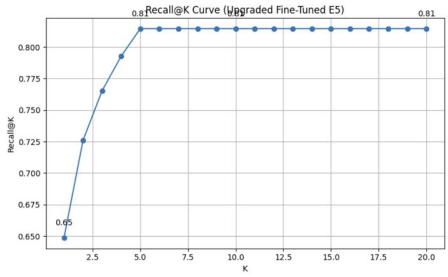

Figure 13: Recall curve (upgraded fine-tuned E5 v2)

7. *Confidence and Similarity Analysis*

To better understand model behavior, we analyze the relationship between similarity-based confidence scores and prediction accuracy.

Results show a strong positive correlation between confidence and correctness. Predictions with higher cosine similarity [5] scores are significantly more likely to be correct.

Accuracy increases steadily across confidence bins, reaching over 90% accuracy in the highest confidence range.

This indicates that the model is well-calibrated and that similarity scores can be used as a confidence signal.

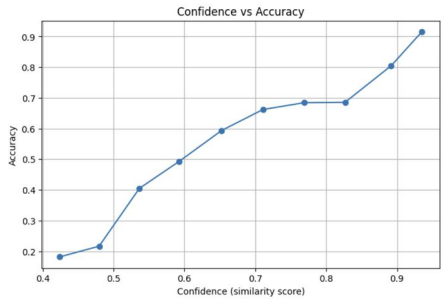

Figure 14: Confidence vs accuracy across similarity bins

The similarity score distribution shows a clear separation between correct and incorrect predictions:

- Correct predictions tend to have higher similarity scores.
- Incorrect predictions are concentrated in lower score regions.
- However, there is some overlap, indicating difficult cases.

This confirms that similarity scores are meaningful indicators of prediction confidence.

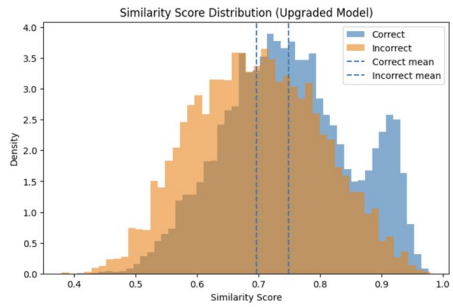

Figure 15: Distribution of cosine similarity [5] scores

8. *Score Distribution Comparison*

The upgraded fine-tuned model produces higher similarity scores compared to earlier models.

This indicates improved embedding alignment, stronger separation between correct and incorrect matches, better ranking confidence.

The shift in distribution confirms the effectiveness of fine-tuning.

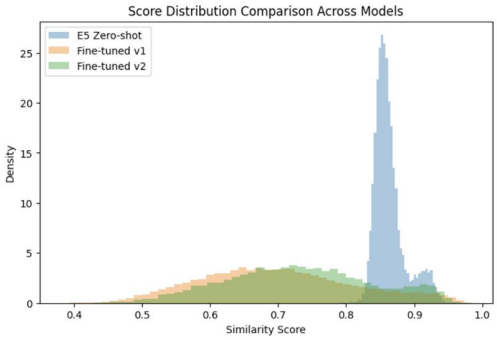

Figure 16: Score Distribution Comparison

9. *Abalation study*

The ablation results show that:

- Replacing MiniLM with zero-shot E5 yields a modest gain.
- Fine-tuning produces the largest improvement by far.
- The upgraded training strategy provides an additional performance boost.

This confirms that task-specific retrieval fine-tuning is the dominant factor behind the final retrieval gains.

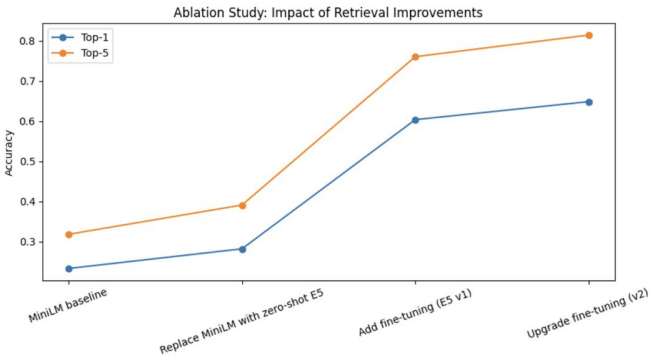

Figure 17: Ablation study: Impact of Retrieval Improvement

10. *Error Analysis*

Error analysis reveals several key failure modes:

(1) Semantic mismatch:
Some Arabic inputs describe concepts that do not map cleanly to a single English idiom, resulting in approximate predictions.

(2) Surface similarity bias:
The model sometimes favors idioms with overlapping lexical patterns rather than true semantic alignment.

(3) Ranking limitations:
In many cases, the correct idiom appears within the Top-5 predictions but is not ranked first, indicating limitations in fine-grained ranking.

(4) Cross-lingual ambiguity:
Certain Arabic expressions have multiple valid interpretations, making it difficult to map to a single idiomatic equivalent.

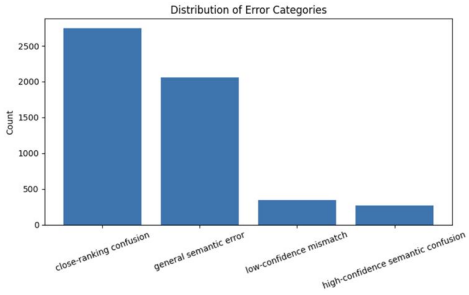

Figure 18: Error distribution across prediction

These findings suggest that while retrieval models capture general semantic similarity, they may struggle with precise idiomatic alignment, particularly in cross-lingual settings.

11. *Qualitative Cross-Lingual Examples*

Qualitative examples demonstrate that the model is able to retrieve semantically appropriate idioms in many cases. For instance, Arabic expressions describing inevitability or shared emotional burden are correctly mapped to idioms such as “come to an end” and “a problem shared is a problem halved.”

However, errors often occur when the input is ambiguous or when multiple idioms share similar meanings, highlighting the difficulty of selecting a single correct expression.

To illustrate model behavior, we present representative examples of Arabic input queries and their retrieved idioms.

Example 1:
Arabic: "انتهى كل شيء في النهاية"
Prediction: "come to an end"
This example demonstrates correct semantic alignment between the Arabic description of inevitability and the corresponding idiomatic expression.

Example 2:
Arabic: "شارك همومك مع الآخرين لتخفيفها"
Prediction: "a problem shared is a problem halved"
The model successfully captures the underlying meaning of shared emotional burden.

Example 3:
Arabic: "لا توجد أي فرصة لحدوث ذلك"
Prediction: "not a chance"
This example shows accurate mapping of strong negation to an idiomatic expression.

Example 4 (Error Case):
Arabic: "يجب أن تبطئ كلامك حتى يفهمك الجميع"
Prediction: "please speak more slowly"
Target: "slow down"
Here, the model retrieves a semantically related phrase rather than the canonical idiom, illustrating surface similarity bias.

12. *Discussion*

The results indicate that Arabic context-to-idiom retrieval is substantially more challenging than its monolingual counterpart.

Unlike direct generation tasks, retrieval-based approaches rely on the quality of the embedding space to bridge semantic gaps between languages. While multilingual models improve alignment, they are still sensitive to subtle semantic differences and cultural variations in idiomatic usage.

Overall, this task highlights the limitations of purely retrieval-based approaches in cross-lingual idiom understanding and motivates the need for hybrid or generative methods that can better capture semantic nuance.

Replacing the MiniLM baseline with multilingual E5 improved retrieval quality substantially. Top-1 accuracy increased from 23.27% to 27.46%, while Top-5 accuracy rose from 31.79% to 38.34%. This suggests that instruction-aware multilingual retrieval embeddings provide better alignment between Arabic contextual queries and canonical English idioms.

13. *Task Summary*

The multilingual E5 model provides the strongest performance for cross-lingual idiom retrieval, outperforming SBERT [5] across all metrics.

However, the moderate accuracy levels and observed error patterns indicate that idiom understanding across languages remains an open challenge, requiring more advanced modeling strategies beyond similarity-based retrieval.

VI. Discussion

The experimental results demonstrate that IdiomX supports a wide range of NLP tasks with meaningful performance.

The experimental findings highlight three key insights. First, idiom detection requires contextual modeling rather than surface-level lexical matching. Second, context-to-idiom generation is not purely a retrieval problem, but requires semantic reasoning. Third, hybrid approaches outperform individual paradigms by combining generative flexibility with retrieval-based grounding.

The experiments suggest that idiom understanding benefits from a division of labor: generative models provide flexible contextual reasoning, while retrieval mechanisms provide discrete grounding and correction.

VII. Applications

The IdiomX dataset supports a wide range of natural language processing tasks, including:

- idiom detection
- idiom generation
- cross-lingual idiom mapping
- multilingual figurative language evaluation

The dataset is particularly valuable for evaluating transformer-based models and large language models in handling non-compositional language.

VIII. Dataset Availability

The IdiomX dataset is publicly available via GitHub and Hugging Face to support reproducibility and further research. The repository includes the dataset, preprocessing scripts, and documentation for easy integration.

Dataset and Tasks links:

https://github.com/aymanshar/idiomx-dataset
https://github.com/aymanshar/IdiomX

IX. Limitations

While IdiomX provides large-scale coverage and rich annotations, part of the dataset is generated using LLM-based methods, which may introduce minor inconsistencies or stylistic biases. Arabic translations primarily reflect standard written Arabic and may not fully capture dialectal variations.

Future work includes incorporating human annotation and extending the dataset to additional languages.

Some generated idiomatic or literal examples may reflect model-preferred phrasing rather than naturally occurring corpus frequency.

Additionally, evaluation for the generation task relies primarily on exact match metrics, which may not fully capture semantic equivalence between idioms.

X. Conclusion

In this work, we introduced IdiomX, a large-scale bilingual dataset for idiomatic expression understanding, providing extensive coverage, rich semantic annotations, and near-complete English–Arabic alignment.

Through comprehensive benchmarking, we demonstrated that idiom understanding requires deep contextual modeling. Transformer-based models significantly outperform lexical approaches on idiom detection, while hybrid generation–retrieval methods achieve the best performance on context-to-idiom prediction.

For the cross-lingual Arabic context-to-idiom task, we showed that retrieval-based models can capture general semantic alignment but remain limited in precise idiomatic mapping. Fine-tuning substantially improves performance, yet challenges such as semantic ambiguity, ranking limitations, and cross-lingual variation persist.

These results highlight that idiom understanding is inherently complex and cannot be fully addressed by a single modeling paradigm. Future work should explore hybrid architectures that integrate retrieval, generation, and cross-lingual reasoning to better capture idiomatic meaning across languages.

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